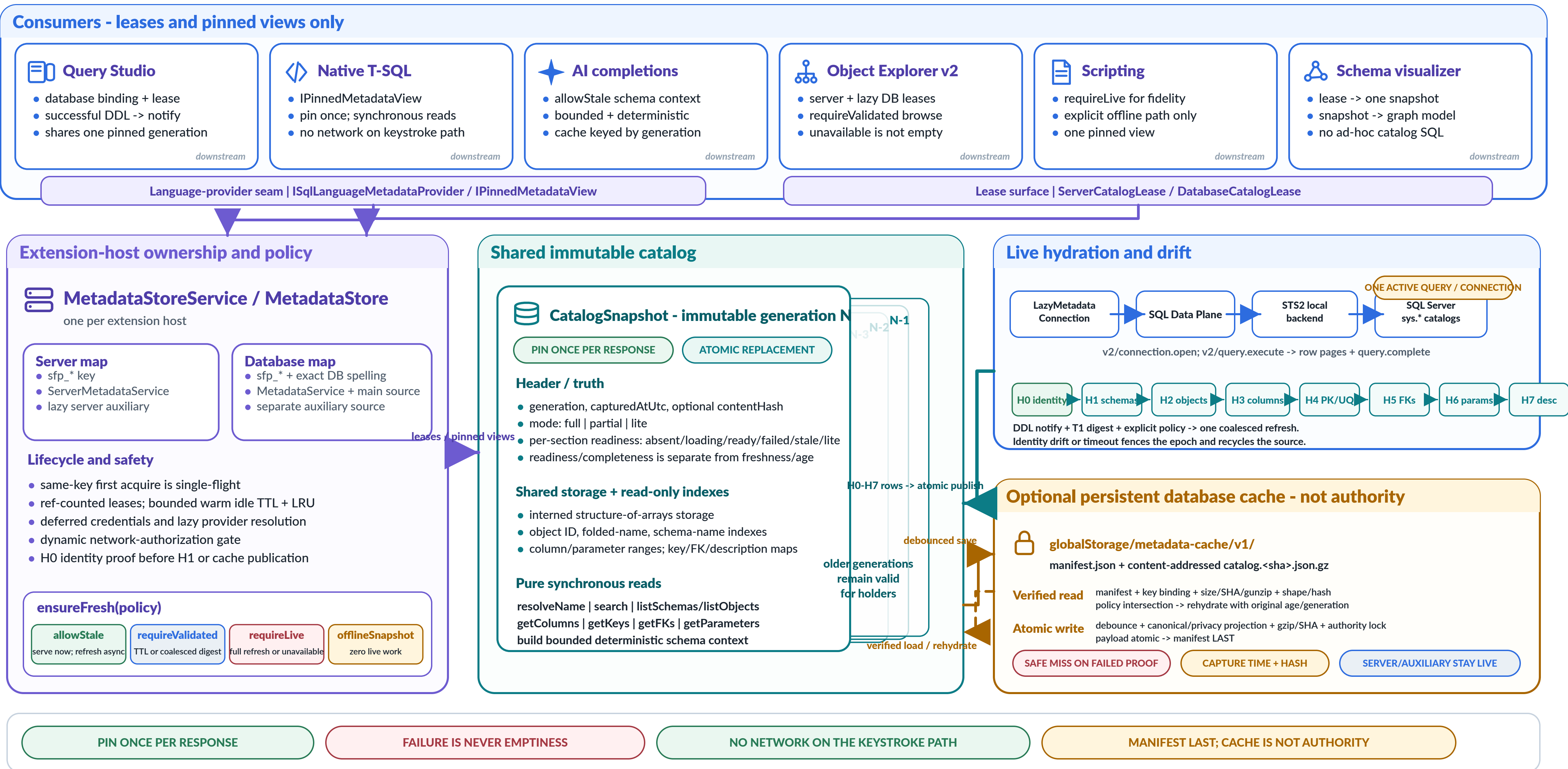
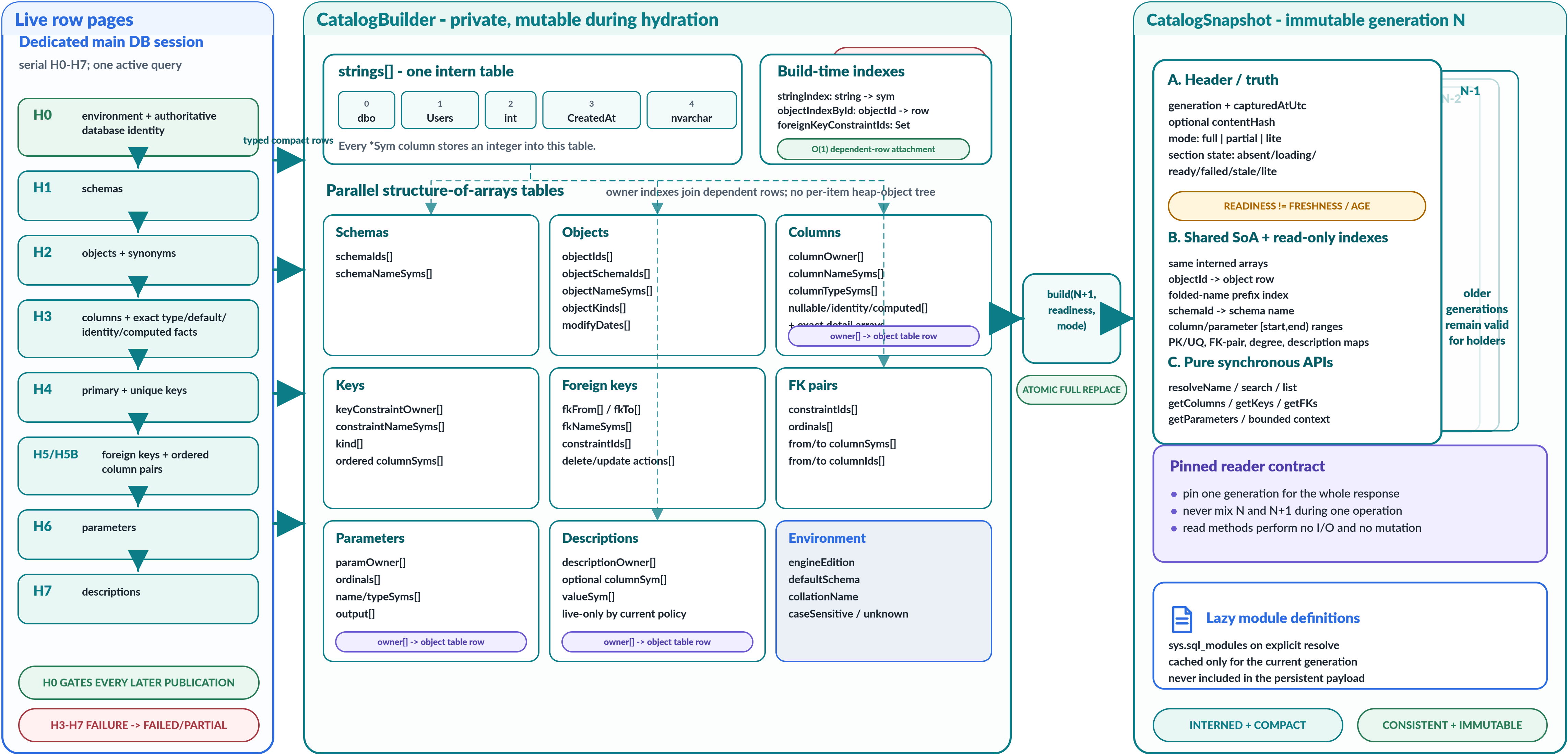


Composite view: PR #22836 substrate below; downstream dev/refactor consumers above (not cut over by this PR).



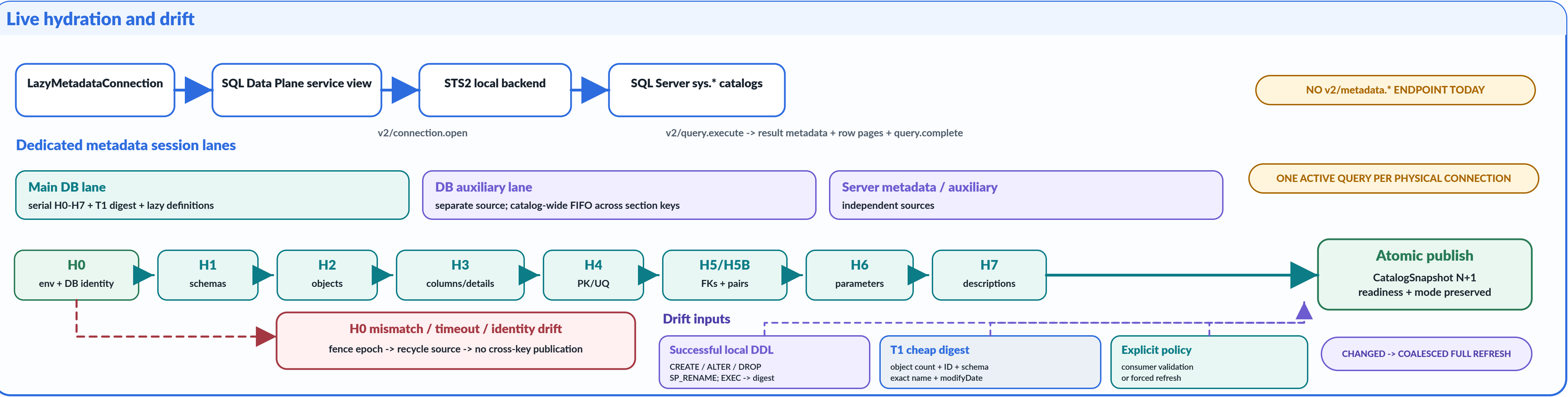
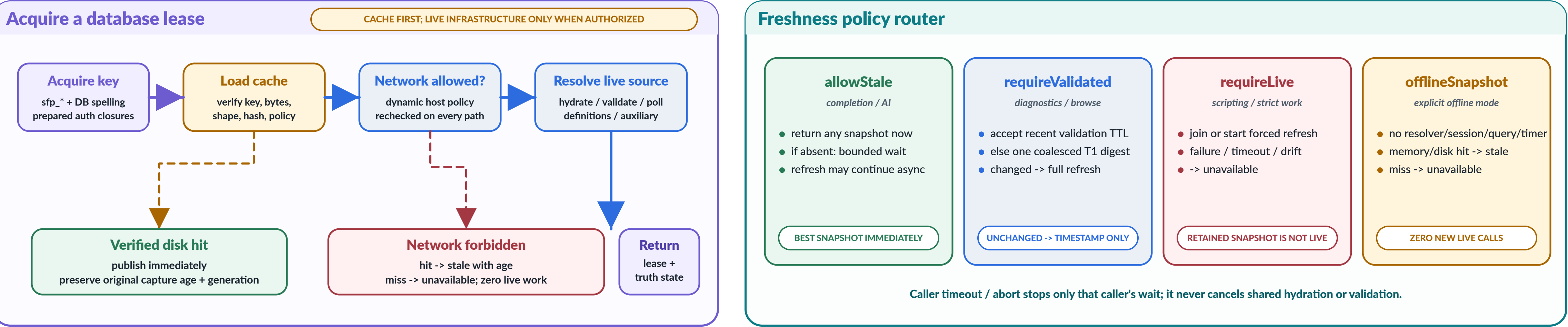
Immutable Catalog Data Layout

SQL catalog row pages are accumulated in a private structure-of-arrays builder, then published as one immutable generation.



Metadata Acquisition, Freshness, and Drift

Cache-first acquisition and live refresh share one rule: return the best snapshot with an explicit truth label, never an empty lie.



Persistent Catalog Cache: Trust and Publication Protocol

The cache is an optional projection of a published database snapshot. It is verified local input, never catalog authority.

On-disk representation



globalStorage/metadata-cache/v1/

index.json (advisory listing/LRU; approximate across windows)

databases/<sfp_*>/<dbh_...>/

.publication.lock

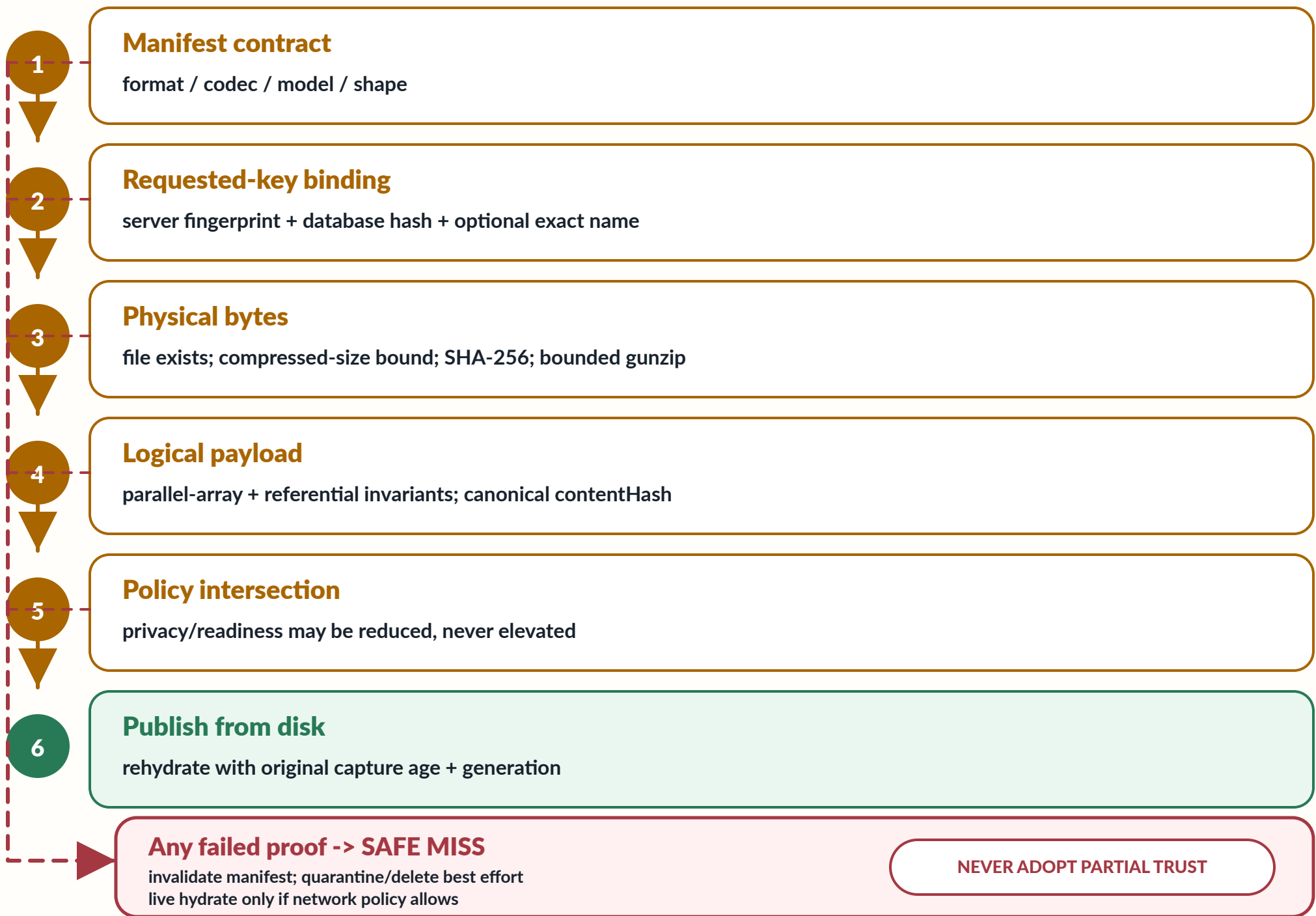
manifest.json

catalog.<compressed-sha256>.json.gz

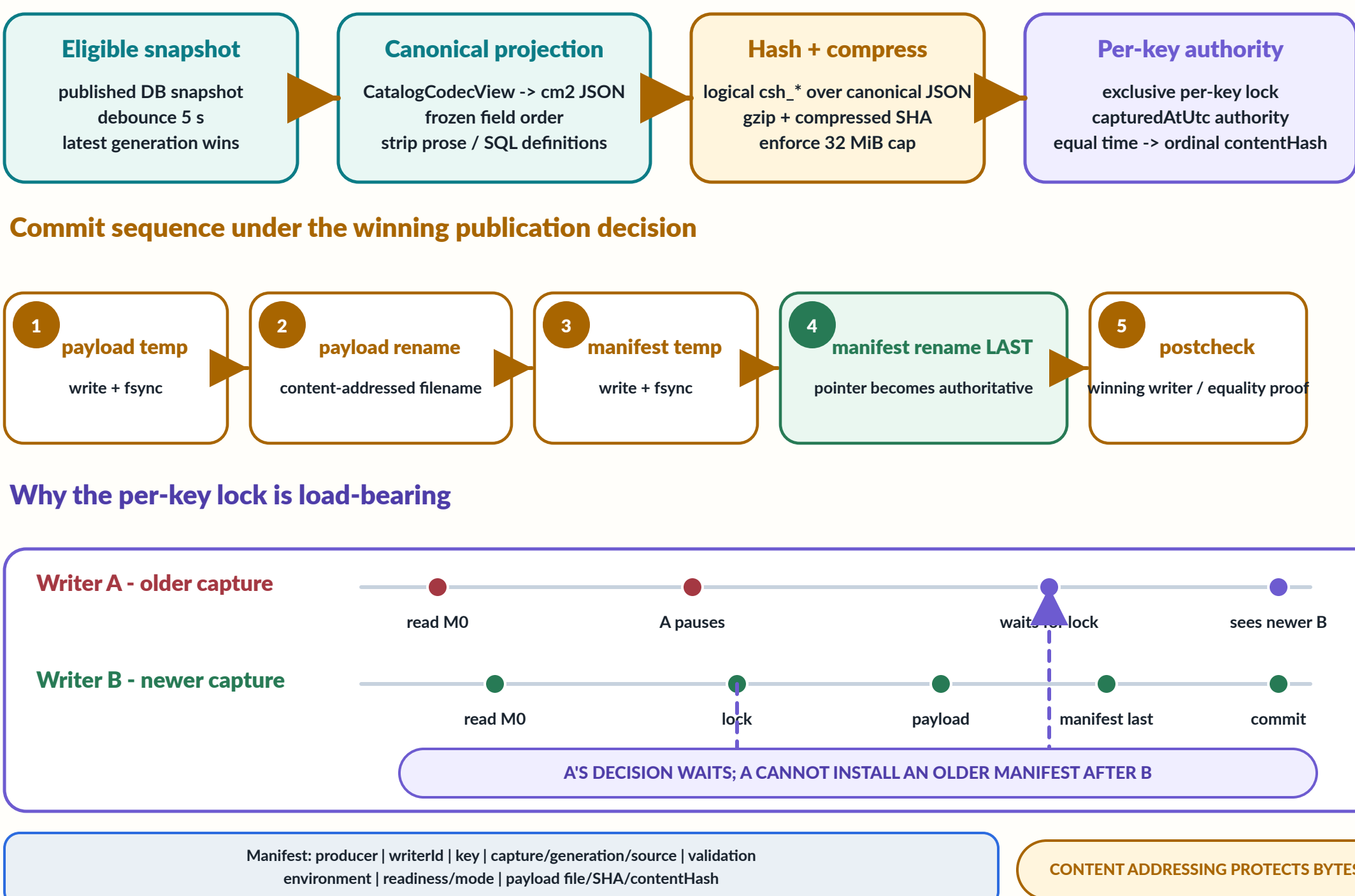
ONLY DATABASE SNAPSHOTS ARE PERSISTED

server catalogs and auxiliary sections remain live/lazy memory structures

Verified READ path



Atomic WRITE and cross-window authority



DEFAULT OFF

MAX AGE 14 DAYS

TOTAL 256 MiB

32 MiB COMPRESSED / ENTRY

ORPHAN GRACE 10 MIN

CLEAR MANIFEST FIRST; REMAINING BYTES ARE INERT